

Summary Table

Feature / Change	Your MOD Already Has	JKK Added Post-Apr 12	Action Needed
Spectrogram FREEZE	✓ Already implemented	✓ Now in upstream	None
Standby mode (<code>g_is_standby</code>)	✗ Removed in your MOD	✓ Still in upstream	Consider re-adding
FPS limiter (45 FPS frame capping)	✗ Not present	✓ New in upstream	Adopt — saves CPU
Dynamic <code>hop_size</code> from ms	✗ Using fixed 21.33ms calc	✓ Uses <code>update_interval_ms = 20</code>	Minor — functionally similar
JSFX: sample-rate change in <code>@sample</code>	✗ Only in <code>@block</code>	✓ Recalculates <code>hop_size</code> in <code>@sample</code>	Adopt — safer for live SR changes
JSFX: <code>a0</code> variable naming conflict	✓ Fixed (uses <code>a0_f</code>)	✗ Still uses <code>a0</code> (shadows LUFS filter <code>a0</code>)	Your fix is better
JSFX: Blackman-Harris window	✓ Implemented	✗ Still Hamming window	Your version is better
JSFX: 24kHz bin limiting	✓ Only processes bins up to 24kHz	✗ Processes all <code>fft_size/2</code> bins	Your version is more efficient
Scope: zero-crossing trigger	✓ Implemented (stable waveform)	✗ Not present (scrolling waveform)	Your version is better

Scope: connected line rendering	✓ Draws line when zoomed in	✗ Always draws bars	Your version is better
Spectrum: dB readout in tooltip	✓ Shows Hz + dB	✗ Only shows Hz	Your version is better
Spectrum: horizontal crosshair	✓ Shows crosshair on hover	✗ Only vertical line	Your version is better
Spectrum: correct log-scale	✓ Proper $\log_{\min_k / \log_{\max_k}} \text{ range}$	✗ Uses $\log(k) / k_{\max_log}$ (DC offset)	Your version is better
Spectrogram: fixed-width canvas	✓ <code>SPEC_RES_W = 800</code> (resolution-independent)	✗ Canvas width = screen pixels (changes with resize)	Your version is better
Spectrogram: cubic interpolation	✓ Uses <code>interpolate_cubic()</code>	✗ Uses linear interpolation	Your version is better
Spectrogram: tilt EQ compensation	✓ 2 dB/octave tilt above 100Hz	✗ Not present	Your version is better
Spectrogram: log-frequency scale	✓ Same log scale as Spectrum	✗ Uses power-law exponent (3.0)	Your version is more accurate
Spectrogram: canvas height rescale	✓ Reblits old data when resizing	✗ Clears canvas on resize	Your version is better
Module width dragging	✓ Drag separators to resize	✗ Fixed ratios only	Your exclusive feature
Color sensitivity	✓ <code>g_color_sensitivity = 1.3</code>	✗ Not present	Your exclusive feature

Gonio: phase bar placement	✓ Reserved space at bottom	✗ Can overlap with content	Your version is better
Font quantization (<code>math.floor + 0.5</code>)	✓ Prevents REAPER font cache overflow	✗ Uses raw float sizes	Your version is better
Theme Presets (4 themes)	✓ Cyberpunk, Neon Green, High Contrast	✗ Not present in Editor	Your exclusive feature
Editor: SaveAllSettings optimization	✓ Only saves on <code>IsItemDeactivatedAfterEdit</code>	✗ Calls <code>SaveAllSettings()</code> on every slider change	Your version is much better
Editor: Spectrogram Gain slider	✓ Has <code>MEM_GAIN_SPECTROGRAM = 9</code>	✗ Not present	Your exclusive feature
Editor: right-click reset on sliders	✓ Right-click resets to 1.0	✗ Not present	Your exclusive feature
Spec floor range	Custom: <code>-90 to -20</code>	JKK: <code>-45 to -45</code> (effectively fixed)	Your range is more useful